

Somnath Jana

Stat-Math Unit, Indian Statistical Institute
8th Mile, Mysore Road, RVCE Post, Bangalore 560 059
rs_math2304@isibang.ac.in | janasomnath364@gmail.com
somnathjanamath.github.io

Education

Indian Statistical Institute — Bangalore, India
Senior Research Fellow in Mathematics — Aug 2025 – Present
Advisor: Dr. Soumyashant Nayak

Indian Statistical Institute — Bangalore, India
Junior Research Fellow in Mathematics — Aug 2023 – Jul 2025
Coursework Grade: A

Indian Institute of Technology Kanpur — Kanpur, India
Master of Science (M.Sc.) in Mathematics — 2021 – 2023
Master's project supervisor: Prof. Parasar Mohanty
Score: 8.5 / 10

Fakir Chand College, University of Calcutta — Diamond Harbour, India
Bachelor of Science in Mathematics (1+1+1) — 2017 – 2020
Score: 742 / 800

Kakdwip Shishu Shikshayatan High School — Kakdwip, India
HSC (10+2 standard) — 2015 – 2017
Score: 441 / 500

Sridharnagar Sailendra Vidyapith — Sridharnagar, India
SSC (10th standard) — 2015
Score: 581 / 700

Research Interests

Operator Algebras, Quantum Information Theory, Matrix Analysis.

Research Articles

- **On the tightness of the Parthasarathy and Schmidt rank bounds for extremal tracial couplings**
  (with Dipankar Dey and Soumyashant Nayak)
  Preprint (2026). hexagon/2610.00489

Fellowships and Achievements

Fellowships & National Qualifications

2023 Selected for the Junior Research Fellowship programme at the Indian Statistical Institute.

2023 Qualified CSIR-UGC NET JRF (Ref: 231610065168 dated 12.10.2023).

2023 Qualified GATE (Mathematics).

2023 Qualified NBHM Doctoral Scholarship Test.

2021 Qualified JAM (Mathematics).

Selective Admissions

2023 Selected for PhD programmes at: IIT Hyderabad and IIT Delhi.

2021 Selected for the M.Sc. programme at IIT Kanpur.

Presentations and Seminars

- Lorentz Spaces and Interpolation *(M.Sc. thesis presentation)*
  IIT Kanpur — Apr 2023

Conferences and Workshops

- Workshop on Noncommutative Analysis and Probability — Jan 2026
  IISc Bangalore (Organisers: Prof. T. Bhattacharyya, Prof. J. M. Lindsay)
- Quantum India Bengaluru
  Bengaluru (Organisers: KSTePS, Govt. of Karnataka, and IISc Quantum Technology Initiative)
- Conference on Operator Algebra and Related Topics (COART-25) — Feb 2025
  IIT Bombay (Organisers: Prof. B. K. Das, Prof. S. Roy, Prof. S. Singh)
- Quantum Probability Day at ISI — Jan 2025
  ISI Bangalore (Organiser: Prof. B. V. R. Bhat)
- Advanced Functional Analysis and its Applications (AFAA 2024) — Dec 2024
  IIT Hyderabad (Coordinator: Dr. Tanmoy Paul)
- Noncommutative Mathematics and Applications (NCMA) — Oct 2024
  ISI Bangalore (Organisers: Prof. J. Sarkar, Prof. D. Goswami, Prof. A. Pal)

Teaching Experience

Teaching Assistant — Indian Statistical Institute, Bangalore

Course names link to the official course descriptions.

- Real Analysis I (B.Math I) — Sem I, 2026–27
  Instructor: Ratna Pal
- Complex Analysis (B.Math III) — Sem II, 2025–26
  Instructor: Ratna Pal
- Function Spaces (B.Math III) — Sem I, 2025–26
  Instructor: Sibaprasad Barik
- Optimization and Numerical Methods (B.S.D.S. I) — Sem II, 2024–25
  Instructor: Kaushik Jana (ISI Delhi)
- Function Spaces (B.Math III) — Sem I, 2023–24
  Instructor: Samir Panja

Languages

Bengali (Native) · English (Full Professional Proficiency) · Hindi (Basic)

References

Dr. Soumyashant Nayak
Associate Professor, Stat-Math Unit
Indian Statistical Institute, Bangalore
R. V. College Post, Bangalore 560059
Email: soumyashant@isibang.ac.in
Website: www.isibang.ac.in/~soumyashant/